An infinite family of fixed-cardinality counterexamples to geodesic-probability monotonicity

Unsigned research draft

30 October 2026
Public research draft; not peer reviewed

Abstract

For every integer $n \geq 4$, we construct two up/right paths on the $(n+1) \times (n+1)$ vertex grid, the second obtained from the first by a boundary unwinding move. Exactly three nonendpoint vertices are selected uniformly and assigned weight one; every other nonendpoint vertex has weight zero. All tied maximizing paths count as geodesics. We prove by elementary classification that the number of three-point configurations making the unwound path optimal, minus the corresponding number for the original path, is $-(5n^2 - 27n + 32)/2$. The difference is strictly negative for every $n \geq 4$. Thus the fixed-cardinality analogue of a path-probability ordering fails in arbitrarily large squares. This is *not* a counterexample to the independent Bernoulli all-parameter conjecture: conditioning on the total number of occupied sites changes the probability measure. The result extends a previously documented $n = 4$ example to an analytically proved infinite family. Priority and independent research significance remain unestablished.

1 Setting and statement

Let $V_n = \{0, \dots, n\}^2$ and $W_n = V_n \setminus \{(0,0), (n,n)\}$. Pick S uniformly from the three-element subsets of W_n, and assign weight 1 to its elements and weight 0 to the other nonendpoint vertices. Since every up/right path contains the two endpoints, arbitrary common endpoint weights do not affect the maximizing set. A path γ is a geodesic when its score $|S \cap \gamma|$ is at least that of every up/right path; all ties are counted.

Let R and U denote right and up steps. For $n \geq 4$ define

$$\gamma_n = \mathrm{RU}^2\mathrm{RU}^{n-2}\mathrm{R}^{n-2}, \qquad \pi_n = \mathrm{U}^2\mathrm{R}^2\mathrm{U}^{n-2}\mathrm{R}^{n-2}.$$

Write $C_\eta(n)$ for the number of three-element sets $S \subset W_n$ for which η is a geodesic.

Theorem 1. *For every integer $n \geq 4$, π_n is obtained from γ_n by a genuine boundary unwinding move as in [1, Section 2], and*

$$C_{\pi_n}(n) - C_{\gamma_n}(n) = -\frac{5n^2 - 27n + 32}{2} < 0.$$

Consequently,

$$\mathbb{P}(\pi_n \text{ is a geodesic}) - \mathbb{P}(\gamma_n \text{ is a geodesic}) = -\frac{5n^2 - 27n + 32}{2\binom{(n+1)^2-2}{3}} < 0.$$

In the signed-run convention of [1], the source runs are $(-1, 2, -1, n-2, -(n-2))$. Their first two partial sums are $-1, 1$. Taking $l = 0, k = 2$ reverses the actual excursion prefix RUU into UUR; the next R merges with that prefix to give π_n. The move is nonidentity and its

interior excursion is nonempty. In particular, we do not invoke the vacuous adjacent-corner case in the printed definition.

The authors of [1] propose their ordering for independent Bernoulli weights and prove strict monotonicity in a sufficiently small positive parameter range. Our theorem concerns the *conditional* uniform three-point ensemble, which has dependent weights. It does not compare independent Bernoulli probabilities. In particular, a negative contribution from one cardinality layer need not determine the sign after summing over all layers.

The case $n = 4$ was already recorded in our earlier finite computation [3]; it is not presented as a new example here. Our contribution within this draft is the elementary analytic extension to the specified infinite family, subject to the priority limitations in Section 5.

2 Reduction to three-point chain geometry

Equip the grid with the coordinatewise partial order: $(x, y) \le (x', y')$ when $x \le x'$ and $y \le y'$. A directed path can contain a set of vertices exactly when that set is a chain in this order. Consequently, for a three-point set S, the largest path score equals the maximum size of a chain contained in S. This also proves that a path is optimal if and only if its own score equals that chain length. No tie-breaking rule is used.

Introduce the four vertices where the two paths differ:

$$a = (1, 0), \quad b = (1, 1), \quad c = (0, 1), \quad d = (0, 2), \qquad L = \{a, b, c, d\}.$$

The free vertices on γ_n but not π_n are a, b; those on π_n but not γ_n are c, d. Let C be their common nonendpoint path vertices and $D = W_n \setminus (L \cup C)$. Explicitly,

$$C = \{(1, 2)\} \cup \{(2, y) : 2 \le y \le n\} \cup \{(x, n) : 3 \le x \le n - 1\}.$$

All vertices in C are above and to the right of all four vertices in L. The sets L, C, D partition W_n.

For each S put $\delta(S) = \mathbf{1}_{\{\pi_n \text{ is optimal}\}} - \mathbf{1}_{\{\gamma_n \text{ is optimal}\}}$. The desired count difference is $\sum_{|S|=3} \delta(S)$. Classify configurations according to $|S \cap L|$ and, when this equals one, according to $|S \cap C|$. The resulting sums are

$\lvert S \cap L \rvert$	$\lvert S \cap C \rvert$	$\sum \delta(S)$
0	any	0
1	0	$-\binom{n-1}{2}$
1	1	$2 - 2(n-3)^2$
1	2	0
2	any	1
3	0	0

(2.1)

We derive each row below. Adding them gives

$$-\binom{n-1}{2} + 2 - 2(n-3)^2 + 1 = -\frac{5n^2 - 27n + 32}{2}.$$

For $n \ge 4$, $5n^2 - 27n + 32 = (n-4)(5n-7) + 4 > 0$. This proves the numerical and sign assertions in Theorem 1.

3 Proof of the configuration sums

3.1 Zero or three vertices in L

When $S \cap L = \varnothing$, both paths visit the same vertices of S, hence their scores and success indicators agree.

If $S \subseteq L$, all four possible triples have longest chain length two. The first two in the table are optimal only for γ_n, and the last two only for π_n:

S	$\{a,b,c\}$	$\{a,b,d\}$	$\{a,c,d\}$	$\{b,c,d\}$
$\delta(S)$	-1	-1	1	1

Their sum is zero.

3.2 Exactly one vertex in L

Suppose first that $S = \{x, h_1, h_2\}$ with $x \in L$ and $h_1, h_2 \in C$. The two common vertices lie on both paths and the local vertex lies on exactly one. That path collects all three ones and is necessarily optimal; the other collects two and is not. Since two choices of x favor each path, the total difference is zero.

Next suppose $S = \{x, z_1, z_2\}$ with $x \in L$ and $z_1, z_2 \in D$. Only the path containing x can be optimal, and it is optimal exactly when S is a three-point antichain. The counts of unordered pairs $\{z_1, z_2\} \subset D$ satisfying this condition are

x	a	b	c	d
#	0	$(n-2)(n-1)$	0	$\binom{n-1}{2}$

Indeed, points in D incomparable with a are in the single column $(0, y)$, $3 \le y \le n$; points incomparable with c are in the single row $(x, 0)$, $2 \le x \le n$. Each such set is itself a chain. For b, the incomparable set is the union of that column and row, and a pair must take one point from each. For d, the incomparable points comprise rows $y = 0, 1$ with $2 \le x \le n$; an antichain pair takes one from each row with the lower-row point strictly to the right. Thus the difference is

$$0 + \binom{n-1}{2} - 0 - (n-2)(n-1) = -\binom{n-1}{2}.$$

Finally let $S = \{x, h, z\}$ with $x \in L$, $h \in C$, $z \in D$. The path containing x collects two ones, the other collects one, and the former is optimal exactly when the three vertices do not form a chain. For each fixed h, the constant numbers of choices $z \in D$ cancel between the two positive and two negative choices of x. Hence the contribution is

$$\sum_{h \in C} \sum_{z \in D} \left(1_{\{a,h,z\} \text{ chain}} + 1_{\{b,h,z\} \text{ chain}} - 1_{\{c,h,z\} \text{ chain}} - 1_{\{d,h,z\} \text{ chain}}\right). \tag{1}$$

For every $x \in L$ and $h \in C$, $x \le h$. The triple is a chain exactly if $z \le x$, $x \le z \le h$, or $z \ge h$. No vertex of D lies below any $x \in L$, and the $z \ge h$ term is independent of x and cancels in (1). Only the rectangles remain.

For $h = (h_x, h_y) \in C$, define

$$T(h) = \sum_{z \in D} \left(1_{a \le z \le h} + 1_{b \le z \le h} - 1_{c \le z \le h} - 1_{d \le z \le h}\right).$$

In the full grid, before deleting L, C and the endpoints, the signed rectangle count is

$$h_x(h_y + 1) + h_x h_y - (h_x + 1)h_y - (h_x + 1)(h_y - 1) = 2h_x - 2h_y + 1.$$

Each common-path vertex contributes zero to the signed sum; the origin contributes zero, and the terminal endpoint is not in any of these rectangles. The four local vertices have signed contributions $1, 1, -1, -2$, respectively, summing to -1. Removing them therefore yields

$$T(h) = 2(h_x - h_y + 1).$$

Summing over the explicit common-path set gives

$$\sum_{h \in C} T(h) = 2 - (n-3)(n-2) - (n-4)(n-3) = 2 - 2(n-3)^2.$$

This proves the remaining one-local-vertex row.

3.3 Exactly two vertices in L

If one of the two vertices is in $\{a, b\}$ and the other is in $\{c, d\}$, the paths have the same score. It remains to compare $S = \{a, b, z\}$ with $S = \{c, d, z\}$, where $z \in C \cup D$. For $z \in C$ the two configurations both contribute success to the path containing the local pair, so their total contributions cancel.

For $z \in D$, the path containing the local pair collects two ones and is optimal unless the triple is a chain. The unrestricted numbers of choices of z again cancel. Let N_{ab} and N_{cd} be the numbers of $z \in D$ forming chains with $\{a, b\}$ and $\{c, d\}$. All points of D below these local pairs have been excluded. Thus

$$N_{ab} - N_{cd} = \#\{(x, 1) : 2 \le x \le n\} - \#\{(0, y) : 3 \le y \le n\} = (n-1) - (n-2) = 1.$$

This is the last nonzero sum in (2.1) and completes the proof.

4 Independent computational checks

The analytic argument above is finite symbolic counting, not an extrapolation from examples. As a separate check, the supplied `verify_sparse_dp.py` enumerates every three-point set for $n = 4, \ldots, 11$. For each configuration it computes the longest chain using an independently written rank dynamic program on the three occupied points, and scores each path using bit masks. It verifies the total difference and every row of (2.1). It does not enumerate all $2^{(n+1)^2-2}$ environments.

For $n = 5$ the saved `n5_independent_check.json` also compares the complete set of 5,984 three-point environments for all 252 paths, using separate lattice-dynamic-programming and point-chain implementations. The two implementations agree on every path count; two separately written move generators agree on all 560 move witnesses, of which 12 have negative difference. This is finite implementation evidence, not a proof-assistant formalization or a substitute for independent human peer review.

n	C_{γ_n}	C_{π_n}	difference	three-point environments
4	309	307	-2	1,771
5	864	853	-11	5,984
6	2,010	1,985	-25	16,215
7	4,122	4,078	-44	37,820
8	7,707	7,639	-68	79,079

See `verify_sparse_dp.json` for the full $n = 4, \ldots, 11$ replay. To reproduce the replay, run `python verify_sparse_dp.py` from the `fixed_k_followup` directory. Python's standard library is sufficient. Run without `-O`, as assertions are part of the checks.

5 Prior work and limitations

The prior geometric path-order questions and maximizing-path setting are discussed by Alberts and Cator [2, Section 8]. The specific bump/unwind operations and small-parameter Bernoulli result are due to McKeown and Nakajima [1]. The $n = 4$ fixed-three-point counterexample appears in the earlier finite companion [3].

A bounded search for fixed-cardinality geodesic comparisons, the explicit path words and the candidate polynomial did not locate the same infinite-family formula. The searched sources and access limits are recorded in `literature_audit_20261030.txt`. Forward citation coverage and the global literature review are incomplete, so exact priority is not established.

Degree elevation, fixed-cardinality decomposition and longest-chain counting are standard methods; no claim of a new general technique is made.

The proof addresses only the displayed path family and exactly three occupied nonendpoint sites. It does not establish the full Bernoulli conjecture, classify other fixed-cardinality levels, or claim publication-quality significance. This unsigned draft has not been human peer reviewed or formally verified. It is deposited as a follow-up in the companion GitHub repository. No journal or arXiv submission is represented by that deposit.

References

[1] S. McKeown and S. Nakajima, *On the most likely geodesic in last passage percolation*, arXiv:2608.11378v2, 20 September 2026, `https://arxiv.org/html/2608.11378v2`.

[2] T. Alberts and E. Cator, *On the Passage Time Geometry of the Last Passage Percolation Problem*, ALEA **18** (2021), 211–247, `https://alea.impa.br/articles/v18/18-10.pdf`.

[3] Unsigned finite research draft, *An exact all-parameter verification of the bump–unwind path order in Bernoulli last-passage percolation for* $n = 4$, 2026, Section 4.1, `https://github.com/SeraphimTsui/bernoulli-lpp-n4`.